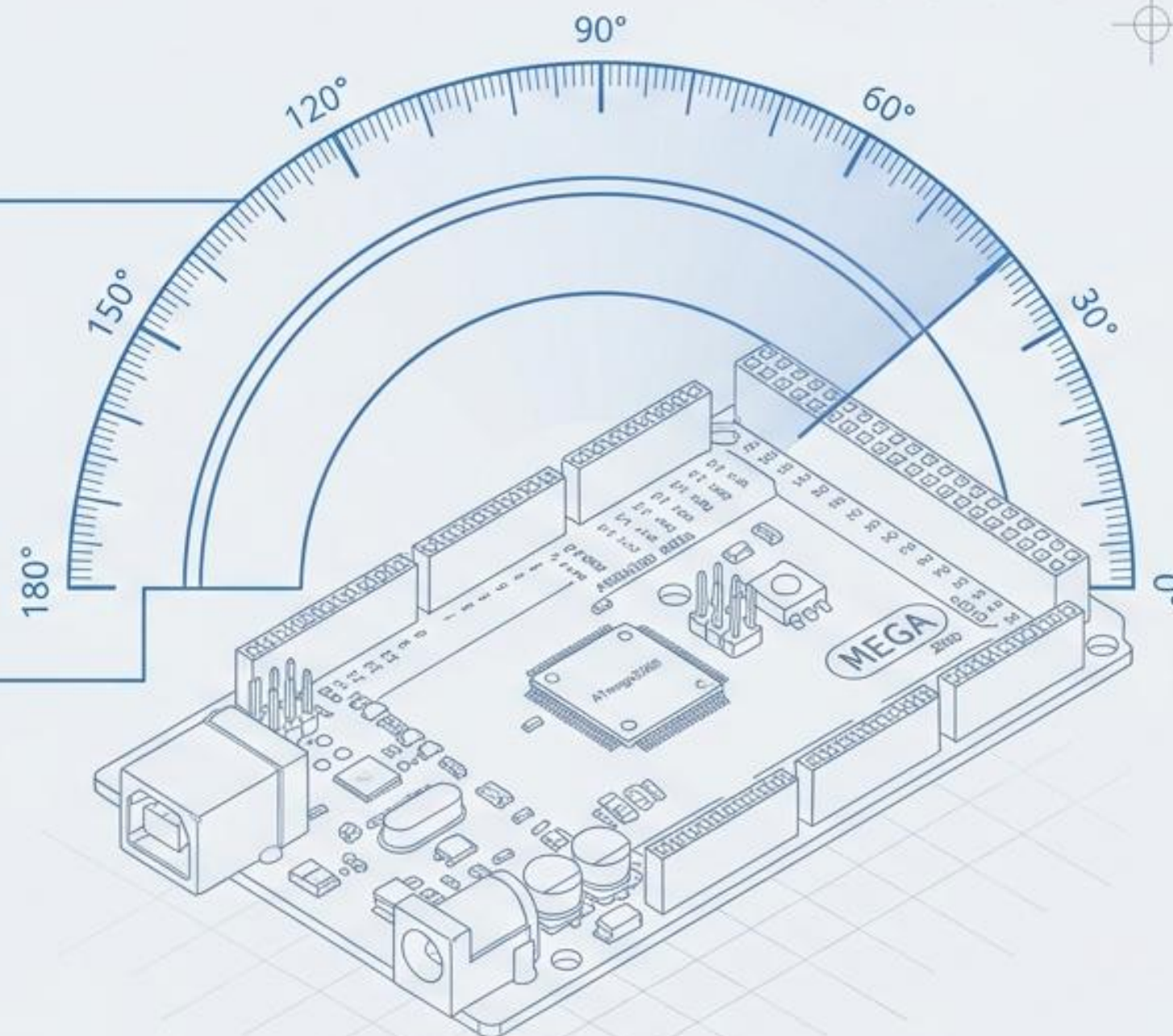




The Smart Radar Platform

A Low-Cost, Fully Integrated Architecture for Real-Time Detection, Tracking, and Visualization



Engineering Team:

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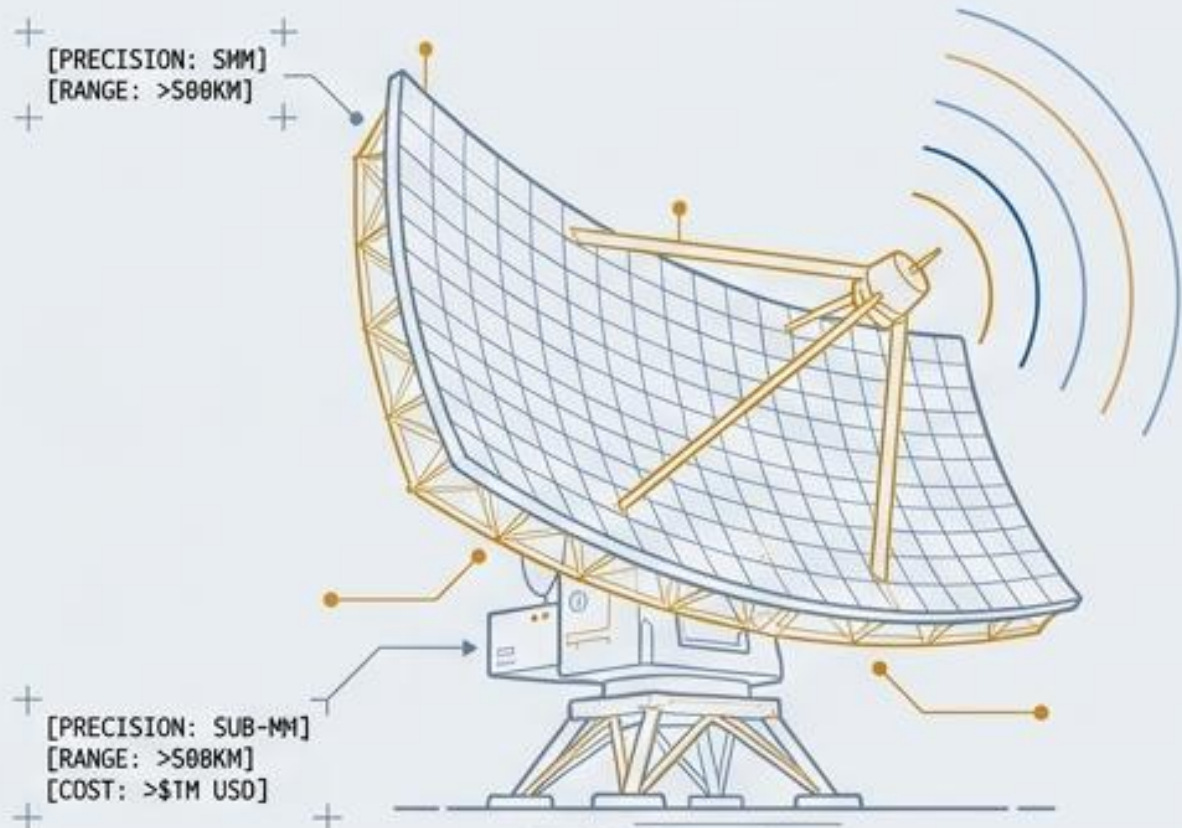
Supervisor:

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Institution:

Ninevah University, College of Electronics Engineering
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The Core Problem: The Radar Dichotomy

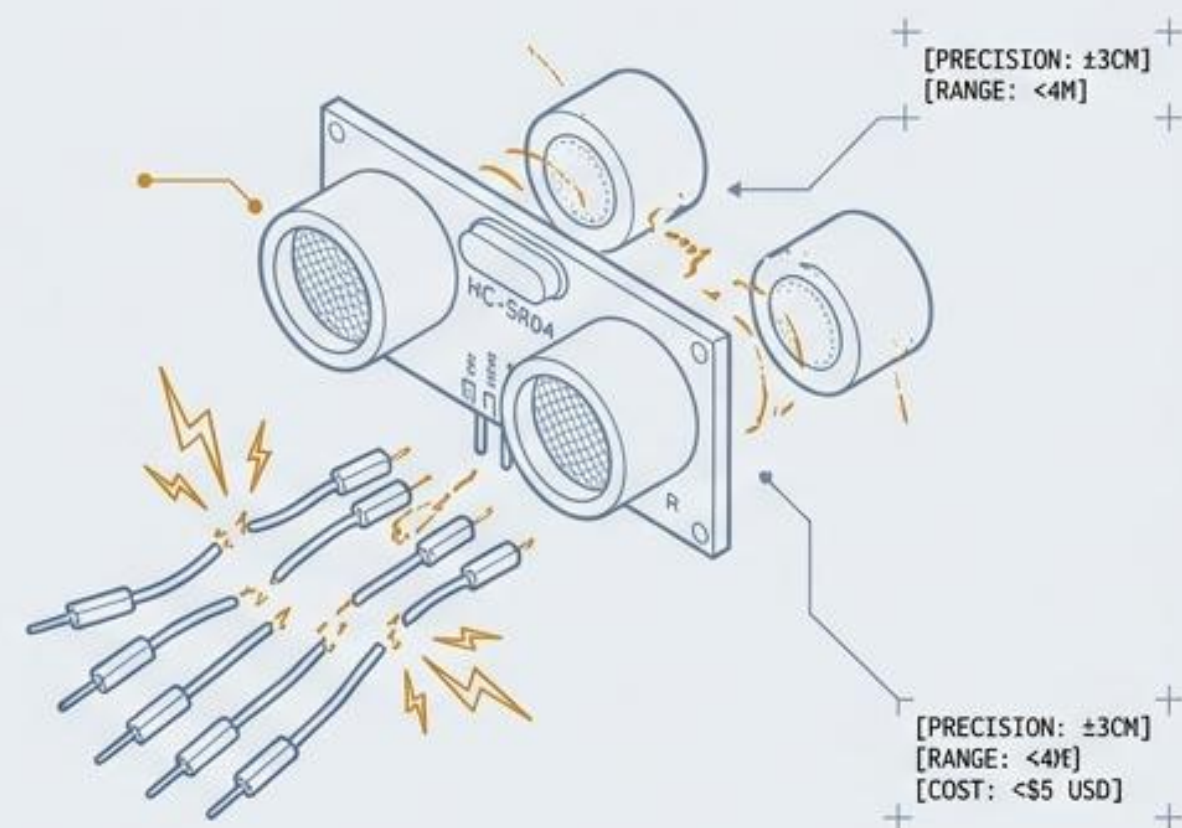


High-End Systems

Expensive, Complex, Proprietary.

Traditional electromagnetic radar provides immense precision but remains financially and technically inaccessible for educational or widespread embedded applications.

The Missing Middle



Basic Ultrasonic Systems

Noisy, Limited, Fragmented.

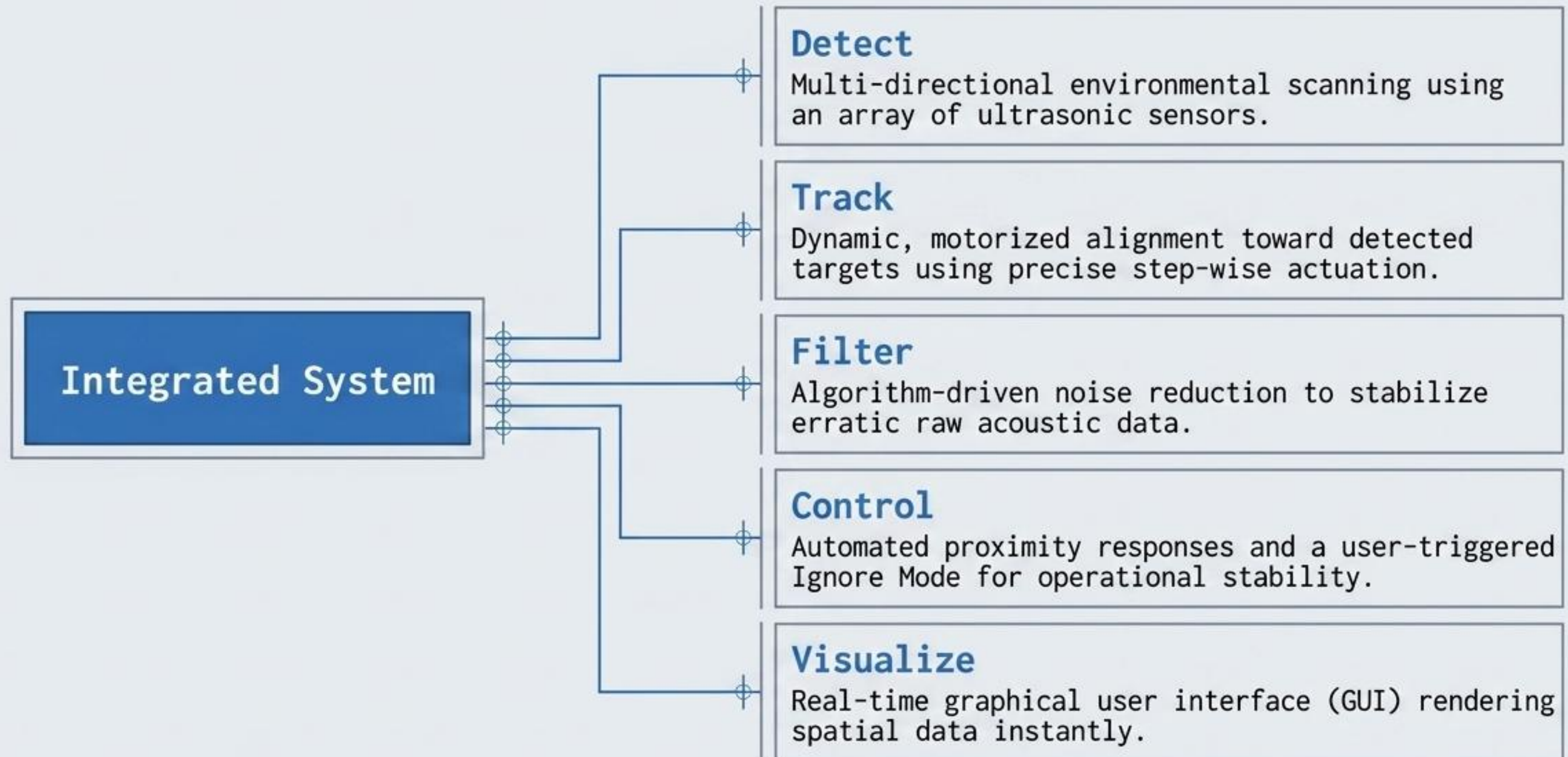
Standard low-cost alternatives lack environmental coverage, suffer from severe blind spots, and lack integrated tracking or visualization interfaces.

The industry requires a **low-cost, unified system** that integrates high-end software **filtering and tracking** with accessible hardware.

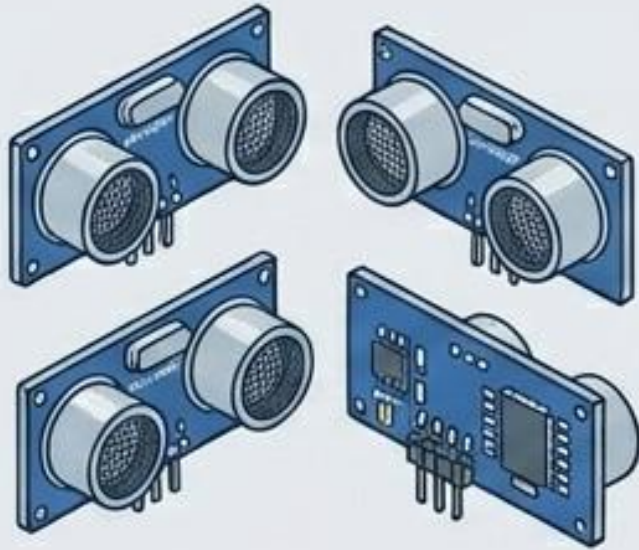
Comparative Analysis: The Integration Gap

System	Wide Area Coverage	Target Tracking Mechanism	Real-Time Visualization	Noise Filtering Algorithm	Proximity Auto-Response
Single-Sensor (Patel et al., 2019)	[X]	[X]	[✓]	[X]	[X]
Multi-Sensor (Kumar & Singh, 2020)	[✓]	[X]	[X]	[X]	[X]
Stepper-Tracking (Rahman et al., 2021)	[X]	[✓]	[X]	[X]	[X]
The Proposed System	[✓]	[✓]	[✓]	[✓]	[✓]

Project Objectives: The Integrated Target State



Hardware Ecosystem: The Physical Blueprint

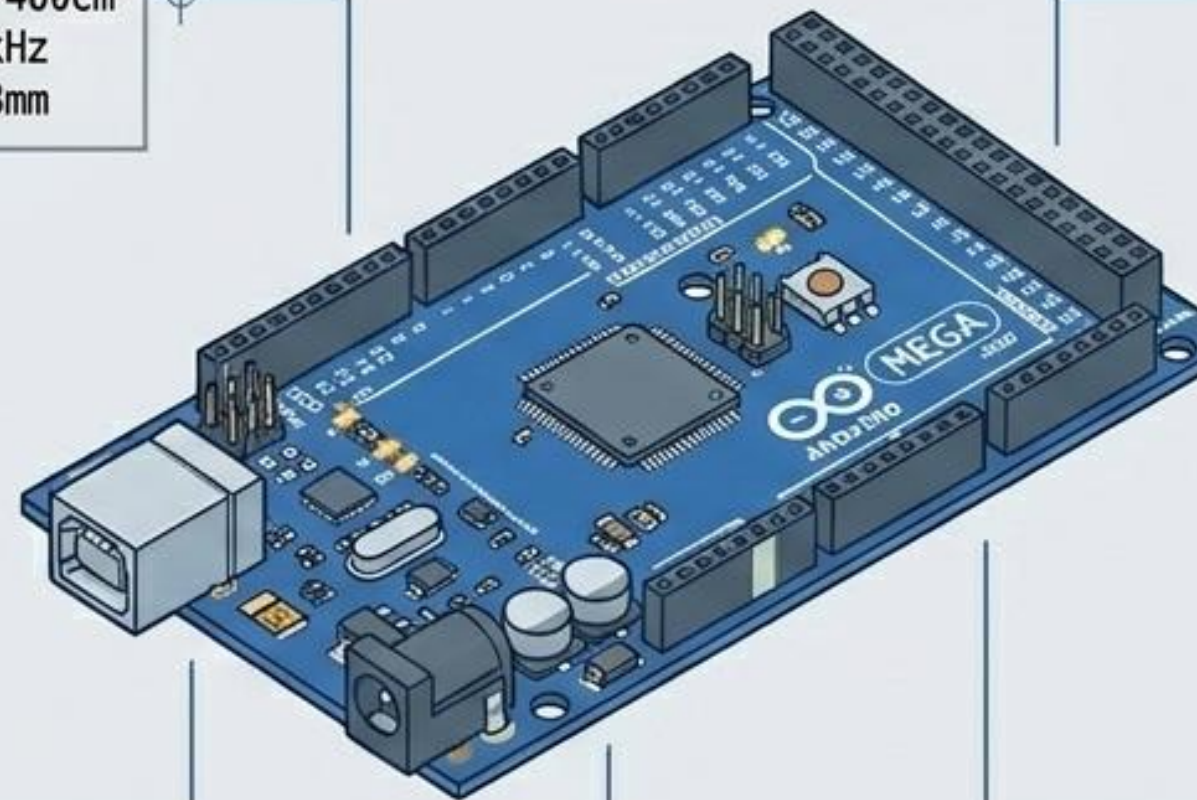


[Sensors]
HC-SR04 x4
Range: 2-400cm
Freq: 40kHz
Acc: +/-3mm

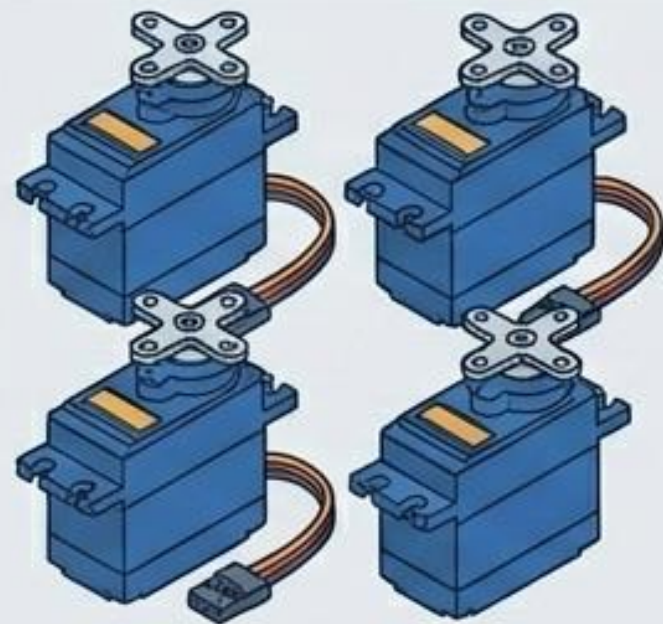


[Alerts]
Active Buzzer | Output: Audible Beep

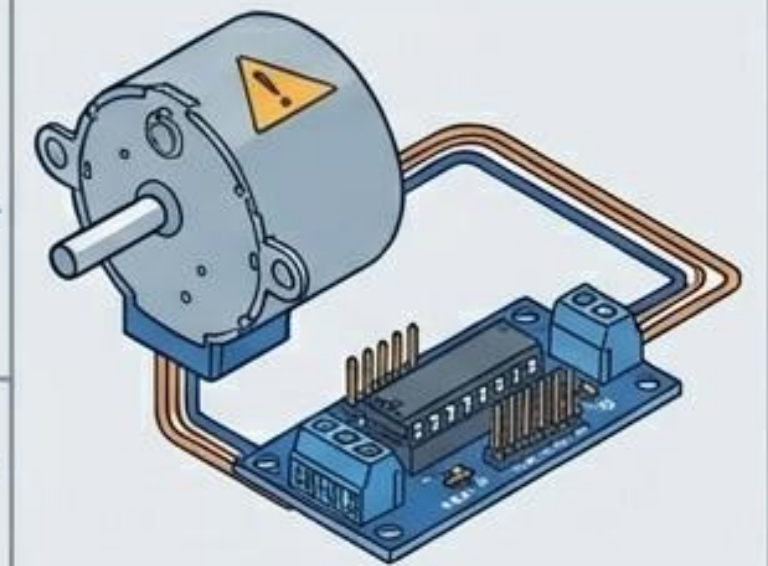
[Power]
12V 4.5Ah source | Regulated 5V logic



Exploded View
ATmega2560 | 16MHz | 54 I/O Pins

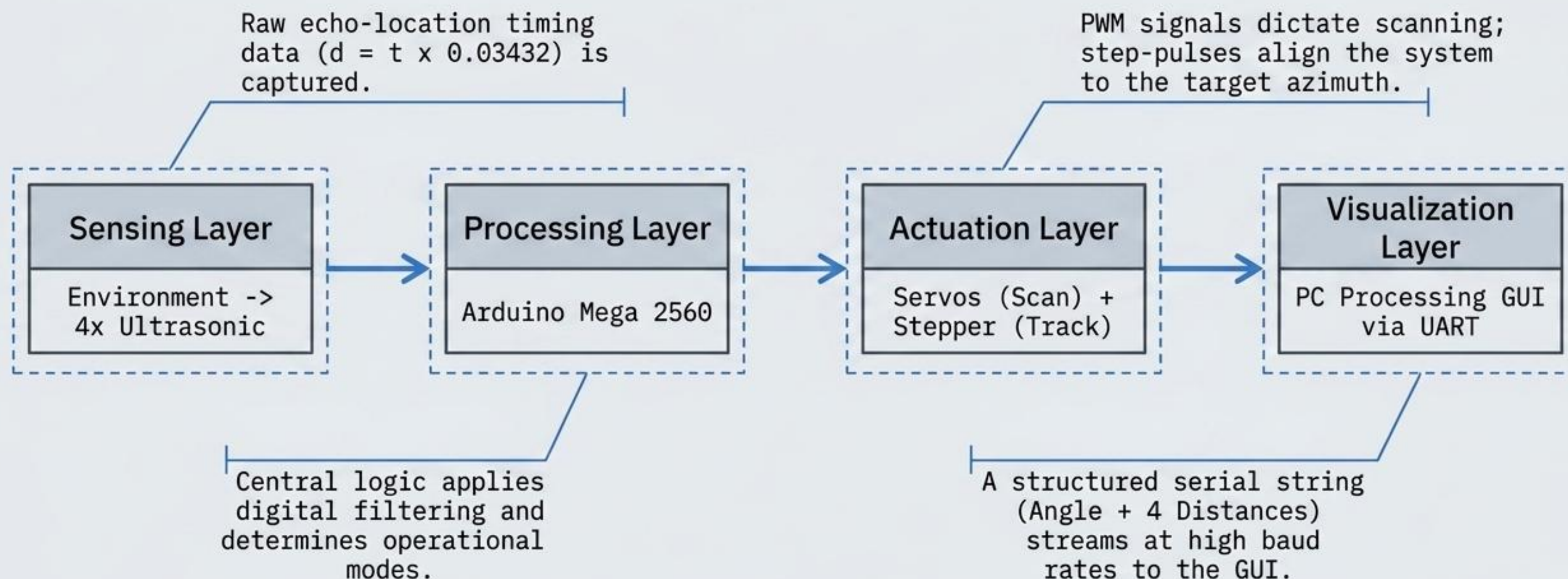


[Scanning]
Servo Motors x4
Range: 0-180 deg
Control: PWM



[Tracking]
Stepper 28BYJ-48
Steps/Rev: 2048
Driver: ULN2003

System Architecture: The Flow of Information



Sensing Strategy: Eliminating Blind Spots

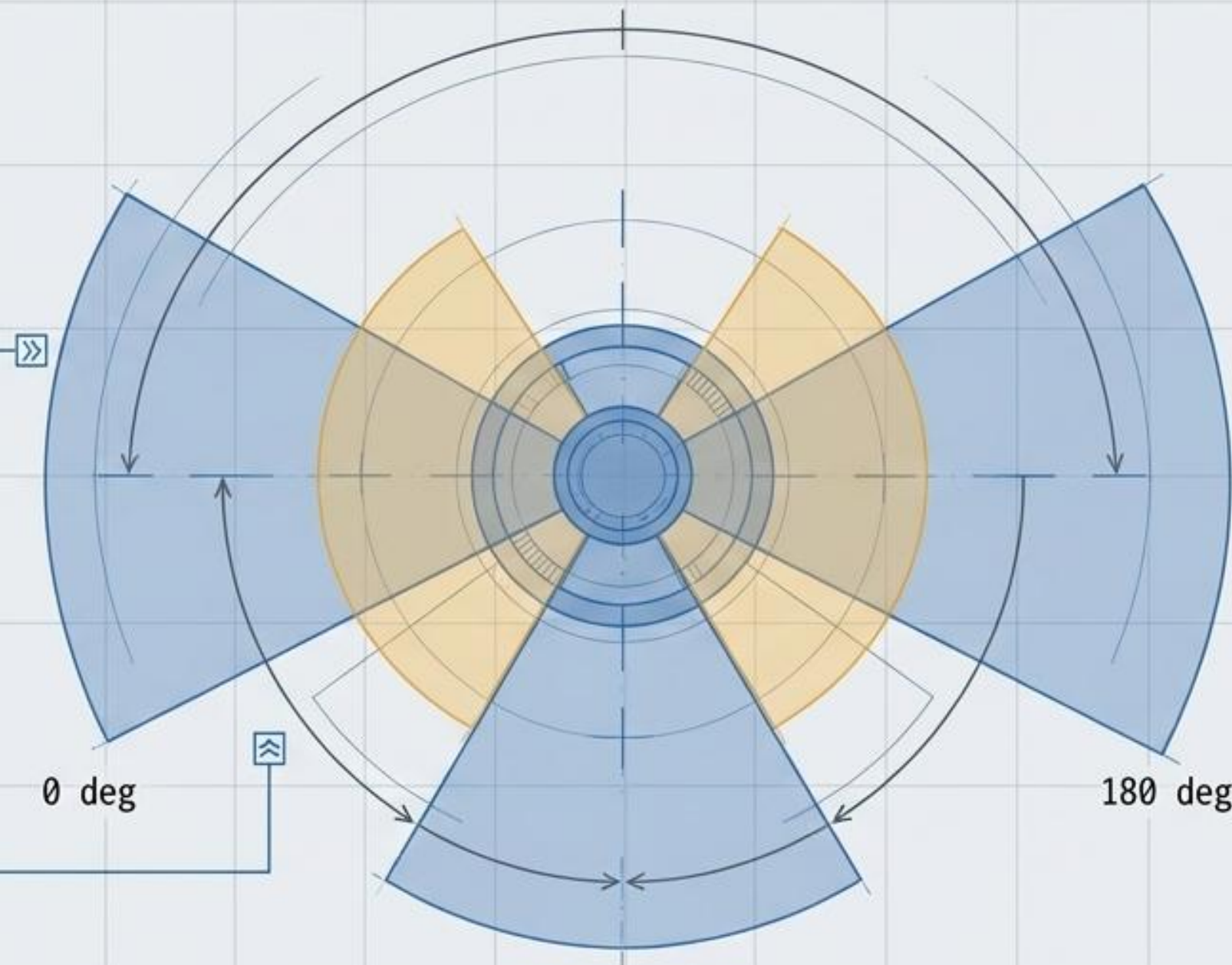
A single sensor creates hazardous blind spots. By architecting a multi-sensor array, the system ensures comprehensive environmental awareness.

Dual-Range Optimization

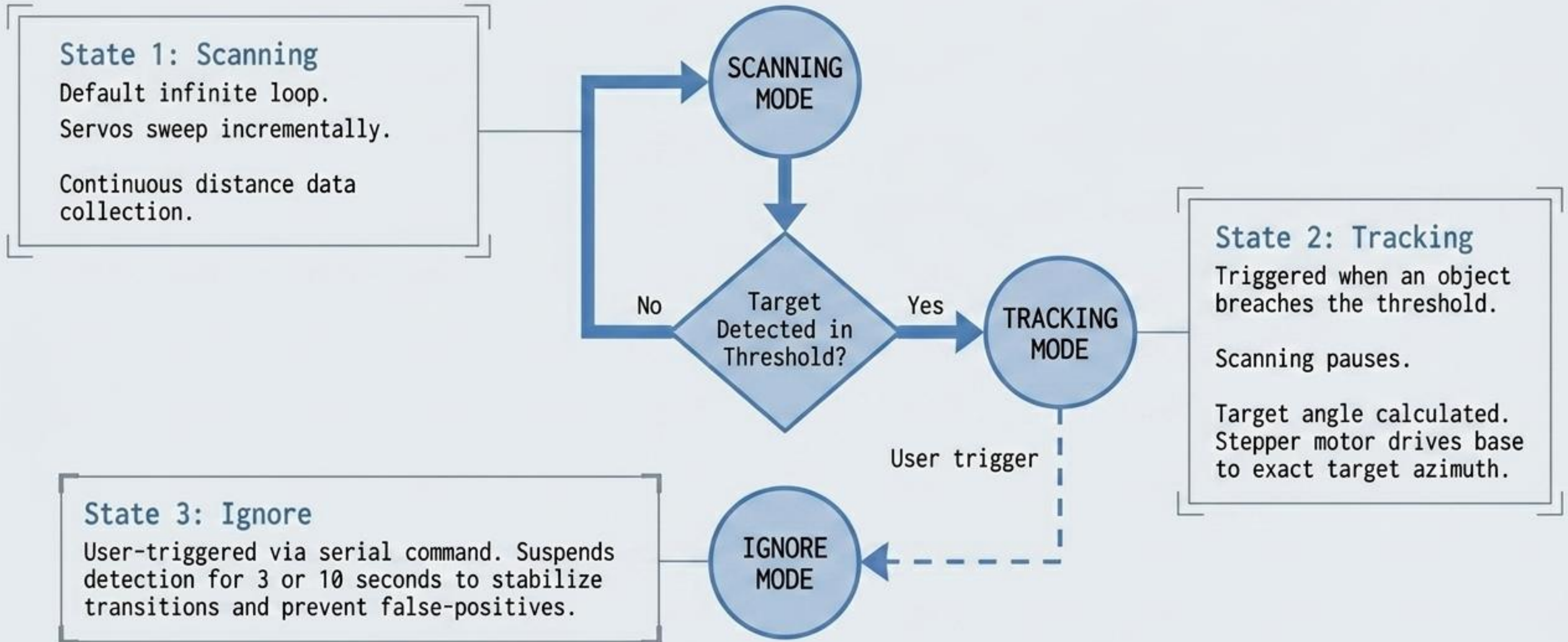
Two sensors are dedicated to long-range environmental mapping, while two are optimized for high-sensitivity close-quarters detection.

Continuous Sweeping

4 independent servo motors synchronize to sweep the entire array across a 180-degree arc seamlessly.

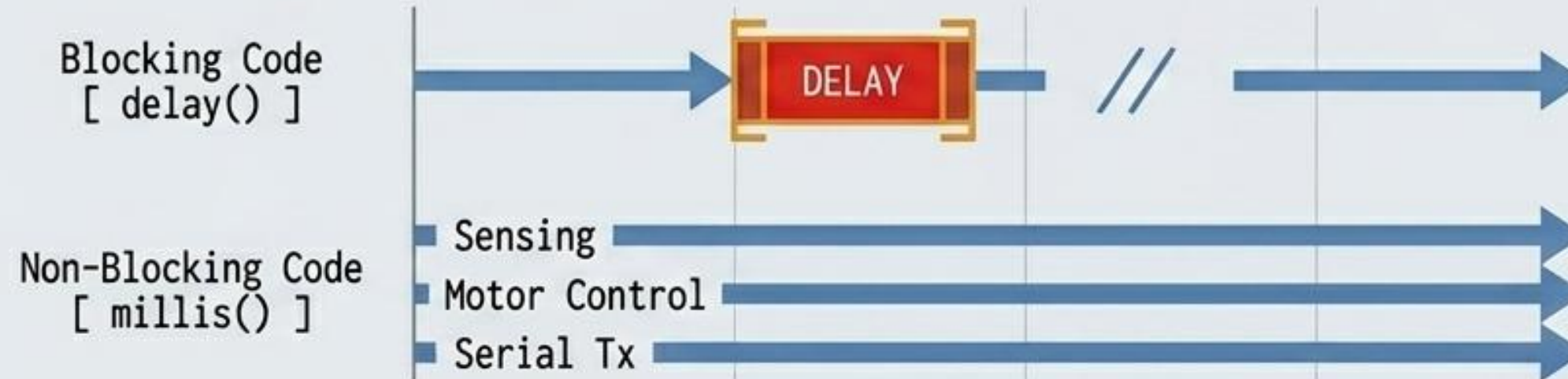


Software Architecture: The System State Machine



Real-Time Processing: Overcoming Hardware Limits

Time Management

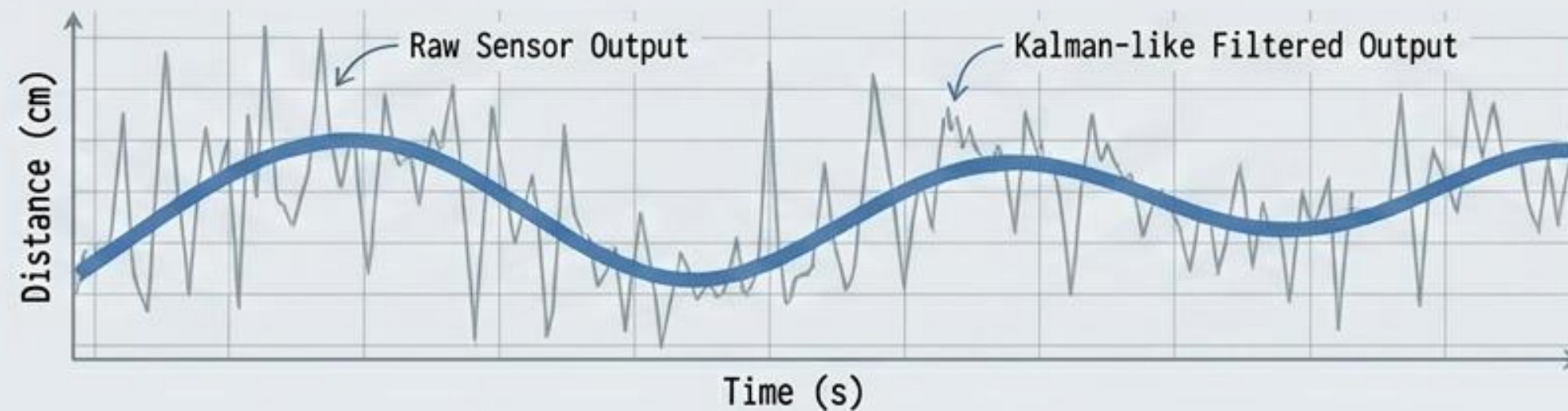


Asynchronous Execution

Traditional delay() functions halt the processor.

Utilizing the millis() function enables true parallel tasking without latency.

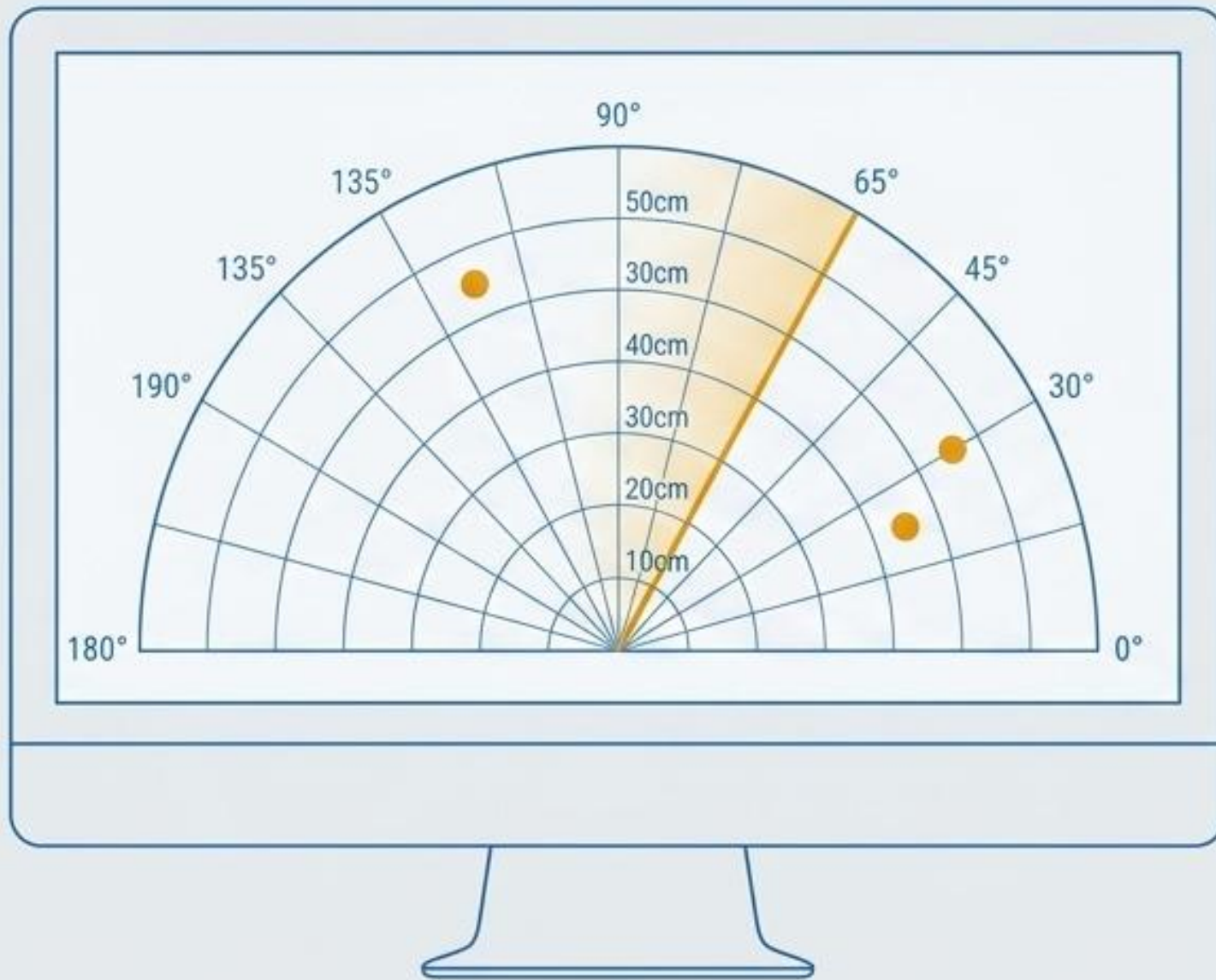
Signal Clarity



Digital Smoothing

Raw ultrasonic data is highly volatile. The system applies a weighted-average digital filter to suppress acoustic noise, enabling industrial-grade stability from hobbyist hardware.

Visualization: The Processing GUI



1. The Data Stream

The [Arduino](#) transmits a continuous structured string via UART:

[Angle, Dist1, Dist2, Dist3, Dist4]

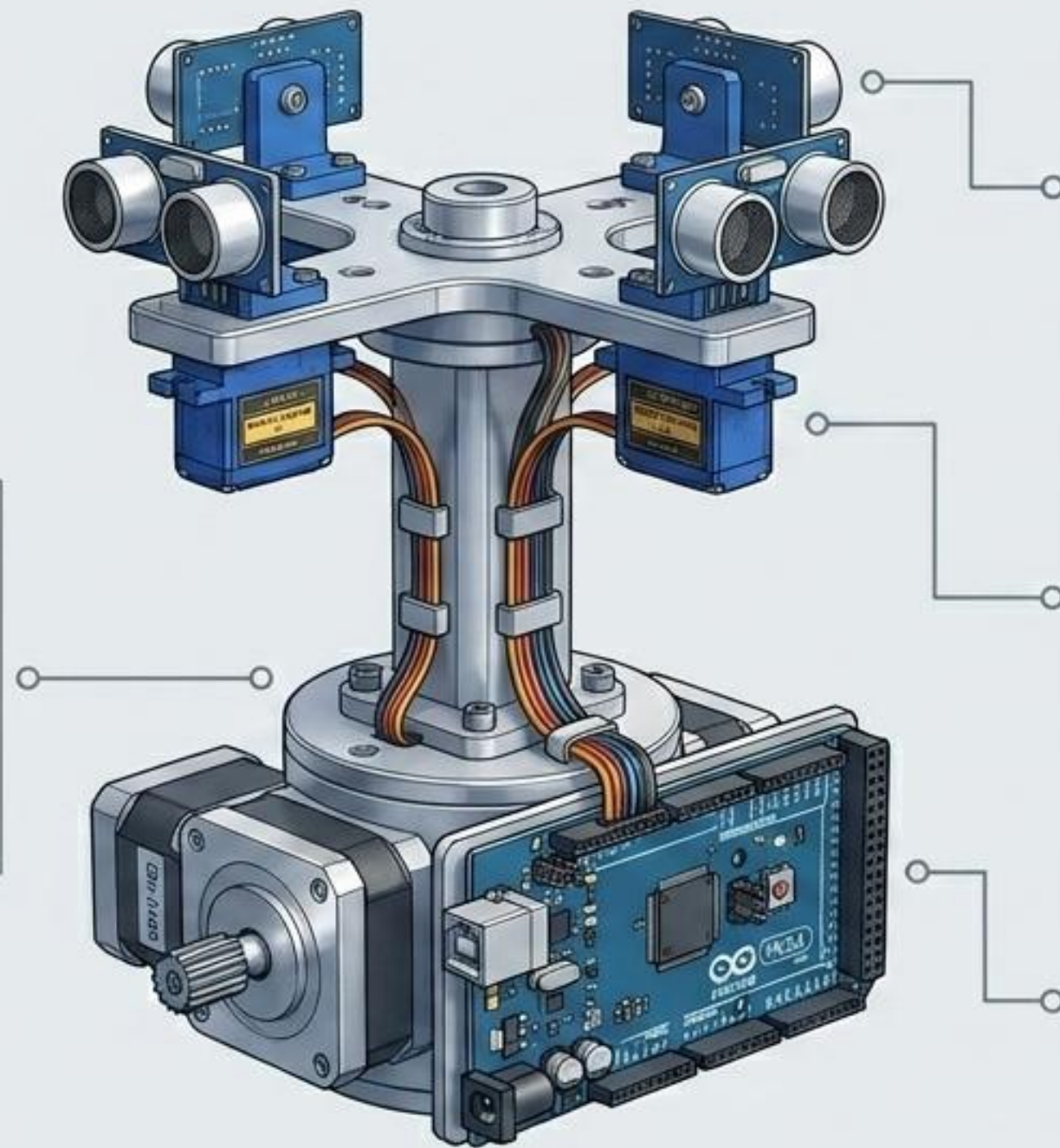
2. Spatial Mapping

Java-based [Processing](#) decodes the serial string to plot distance vectors accurately on a dynamic polar coordinate grid.

3. Target Rendering

Detected objects appear as distinct plotted points instantly, translating raw text into spatial situational awareness.

The Final Prototype: Hardware Integration



1. Quad-Sensor Array (HC-SR04)

Mounted on independent rotational bases.

2. Servo Motor Bank

Driving the 0-180 deg continuous sweep.

Central Processing Hub

Arduino Mega 2560 and distributed 5V/12V power management.

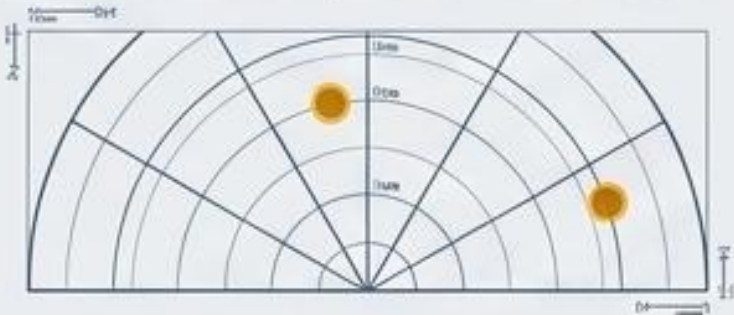
Stepper Base Unit

28BYJ-48 providing precision azimuth alignment for the entire platform.

Performance Validation Dashboard

Multi-Target Integrity

SNS 1 @ 21cm | SNS 3 @ 33cm



Distinct tracking of simultaneous targets without data collision.

Close-Range Sensitivity

SNS 2 @ 6cm



Stable plotting at extremely short ranges despite high echo-risk.

Mid-Range Accuracy

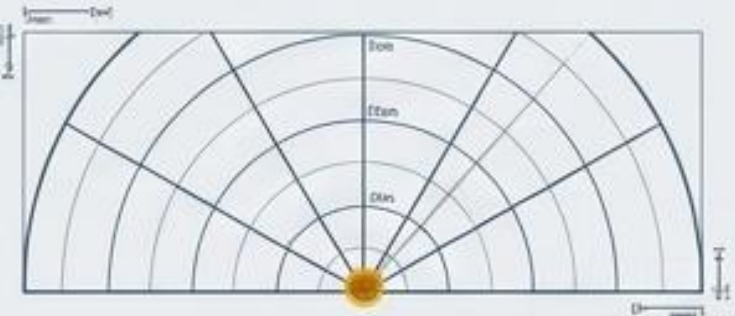
SNS 3 @ 15cm



Consistent positional rendering with zero signal fluctuation.

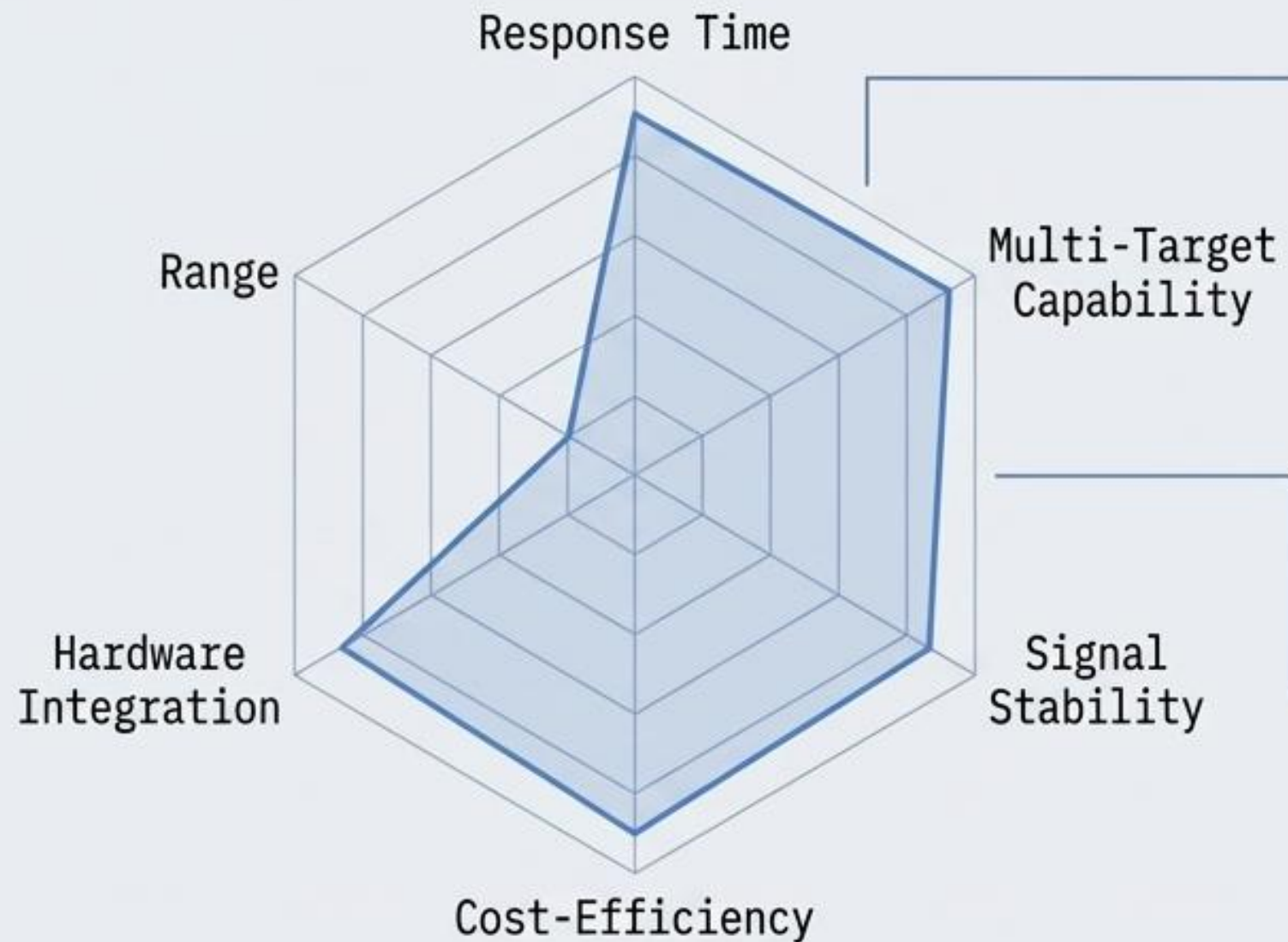
Extreme Proximity Edge-Case

SNS 4 @ 4cm



Maintained stability at physical limits due to digital filtering.

Performance Analysis & Synthesis



System Strengths

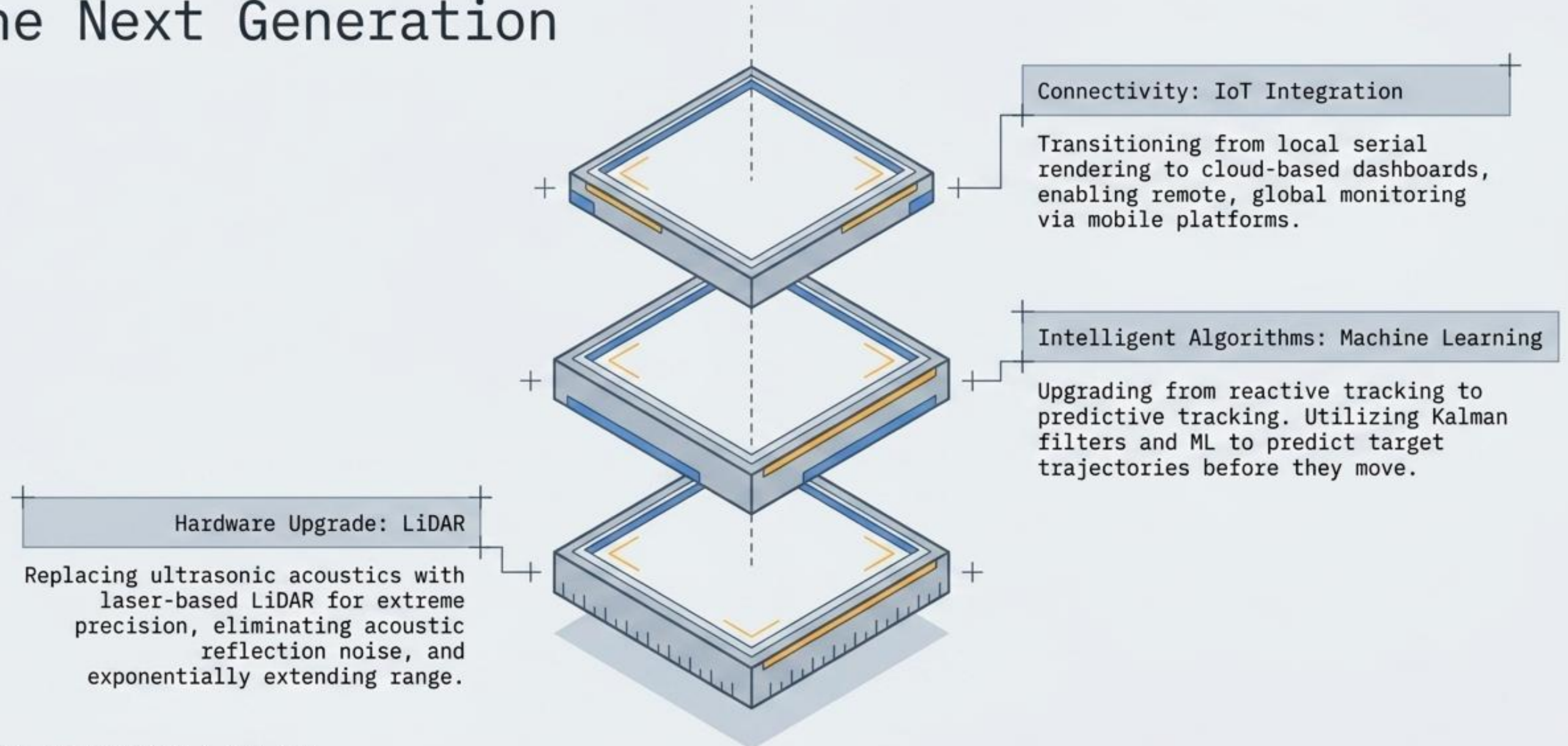
Unprecedented response time and signal stability for this price tier. The non-blocking software architecture ensures the GUI, sensors, and motors remain perfectly synchronized.

Inherent Limitations

Constrained by the physics of ultrasonic waves. The system is susceptible to specific environmental reflections and is limited to short/mid-range operations compared to electromagnetic radar.

Core Insight: Smart software architecture successfully compensates for low-cost hardware limitations, creating a professional-grade embedded system.

Future Evolution: The Next Generation



Mission Complete: System Validated

By unifying multi-sensor arrays, non-blocking software architecture, and real-time GUI rendering, this project **successfully** bridges the gap between low-cost hardware and high-performance radar tracking.

Developed in the Department of Systems and Control Engineering,
Ninevah University.

Special thanks to Dr. Thakwan Akram Jawad for foundational guidance
and technical oversight.